The Next Screenshot Knows: Gated Hindsight Distillation for Mobile GUI Agents

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Paper version: arXiv 2608.06065v1 · CUA Radar Reel poster, not made or endorsed by the authors

Problem

> *the evidence often appears only on the subsequent screen*

Offline prefix-action training throws away the next screenshot — the evidence that explains the demonstrated action. The rationale appears on screen only after the tap.

Why the supervision gap

1. **No why.** Labels say what was tapped, not why.
2. **No evidence.** The prefix often lacks the cue.
3. **Hindsight.** The next screen turns prediction into inference.

Evaluation setup

AndroidWorld · AndroidLab · Qwen2.5-VL-7B · Qwen3-VL-8B · Pass@1, online · 3 runs, mean ± std · 200 steps, G=8 · 420×896 screenshots

- **Eval:** online task success; **training:** offline OpenMobile trajectories.
- **Hard subset:** SFT-fails-once examples (27,360 → 6,968 / 5,982).

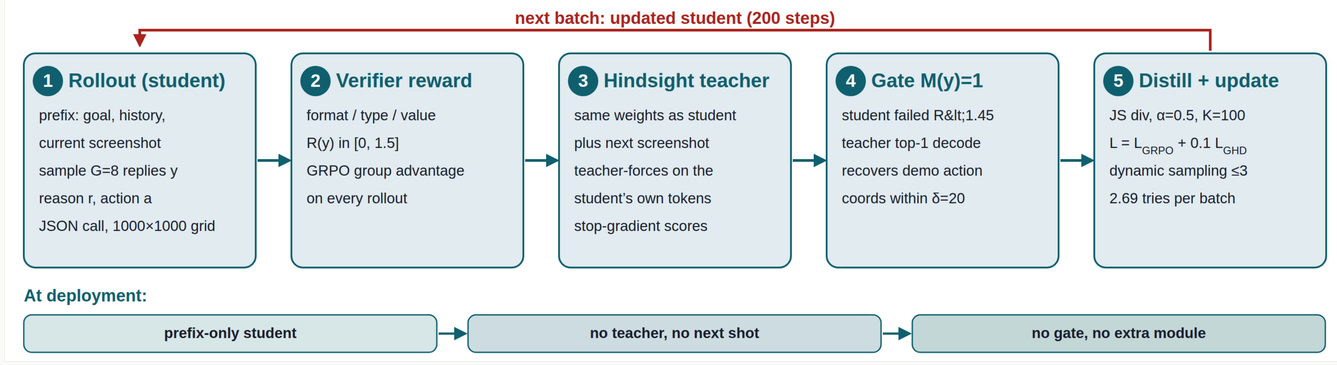
Our read (CUA Radar)

62	★★	online
score of 100	band	evidence level

- **Reports:** online eval; 3-run std; same init/data/budget; ablations + STaR transfer check.
- **Not reported:** held-out apps; full cost; code only announced; bases are 7B/8B only.
- Sub-pattern R01. Authors' numbers.

Method: Gated Hindsight Distillation (training-time only)

- **Sees / acts:** goal + action history + current screenshot in; reasoning then a JSON tool call out (click / swipe / type) on a 1000×1000 grid.
- **Teacher:** the same weights plus the realized next screenshot, re-scoring the student's own rollout tokens.
- **Signal:** rule step reward (type / value / format) drives GRPO; gated token-level distillation only where the student fails and the teacher verifiably corrects it.



Our redraw (not the paper's figure): the GHD training loop; at inference only the prefix-only student remains.

$$L = L_{\text{GRPO}} + \lambda L_{\text{GHD}}, \quad \lambda = 0.1$$

GRPO on every rollout; hindsight distillation only on gated failures

Results: online Pass@1 (%)

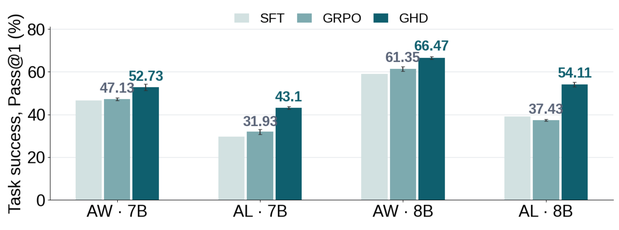


Table 3, redrawn: mean ± std, 3 runs, same data/init.

GHD > GRPO on every model × benchmark pair; eval is online

SO WHAT → *AW = AndroidWorld; AL = AndroidLab.*

Ablation: AndroidWorld Pass@1 (%)

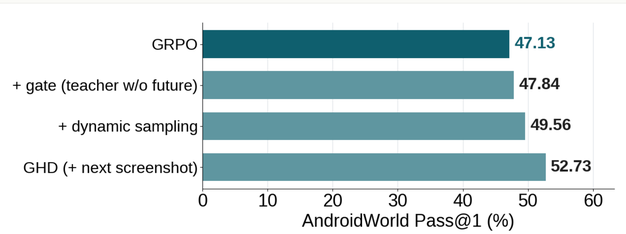


Table 2, redrawn: 7B model, same SFT init; Gate / DS controls get no future information.

SO WHAT → *Gate +0.71 and sampling +2.43 need no future; the next screenshot adds +3.17, the largest step. Distillation also beats a STaR-style transfer under every privileged signal (Fig. 3).*

Headline Numbers

66.47%
AW Pass@1, Qwen3-VL-8B
vs GRPO 61.35 ± 1.08, 3 runs

+5.60 AW pts vs GRPO (7B)
43.10% AL Pass@1, 7B
54.11% AL, 8B

The paper's own controls

CONTROL	RESULT	READ
Gate + DS	+0.71/+2.43	small
Rollout budget	+2.29/+14.50	not sampling
Teacher input	-0.43 vs +5.62	next shot
GUI-Shift aux	+0.28 vs 5.60	distill wins

AW Pass@1, 7B; deltas vs GRPO.

Takeaway

The next screenshot is training-only privilege: a same-weight teacher re-scores the student's failed steps and distills them only where it verifiably recovers the demonstrated action; deployment stays prefix-only, with no extra inference module.

no extra inference cost · 7 of 9 AL apps · 2.69 attempts per batch · τ=1.45, δ=20, K=100

Scan to read the full paper

arxiv.org/abs/2608.06065v1